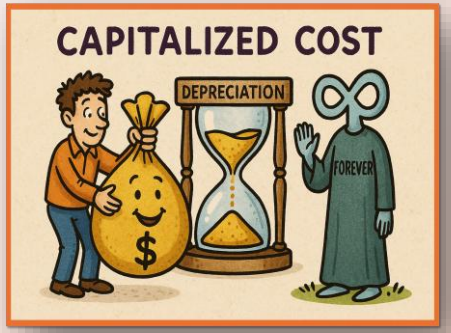
CAPITALIZED COST: Quiz

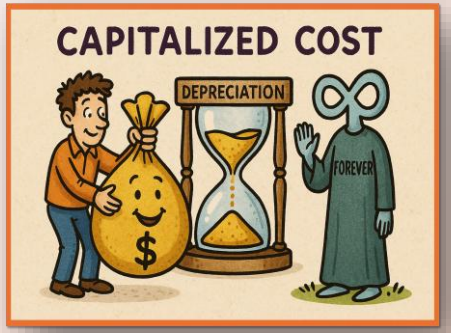


Look for the capitalized cost of a lab site that requires:

- Initial cost ₱5m for construction, ocular inspection at ₱0.1m and tax worth ₱0.3m.
- ₱0.1m at the end of every year for the 1st 6 years and ₱0.120m each year thereafter for operating expenses.
- ₱0.5m salvage cost and every 5 years for replacement of equipment
- Interest at 12% per semi-annual.

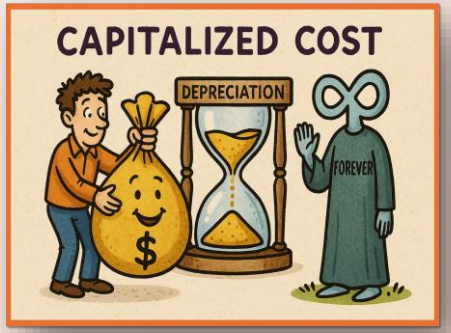
- What do I need to do to own a car?
- What are the expectations/responsibities in owning a car?
- What is the owners' expectation on a new car? What about in 5 years? 10 years?

CAPITALIZED COST



- Specialized application of PERPETUITY.
- Capitalized cost of any property is the summation of:
  - FIRST COST (FC)
  - Present worth of all costs:
    - ✓ REPLACEMENT
    - ✓ OPERATION
    - ✓ MAINTENANCE for LONG TIME or FOREVER/ETERNITY (INFINITY).
- **3 cases have CC.**

CAPITALIZED COST: Case 1



- No replacement, only maintenance and/or operation every period.

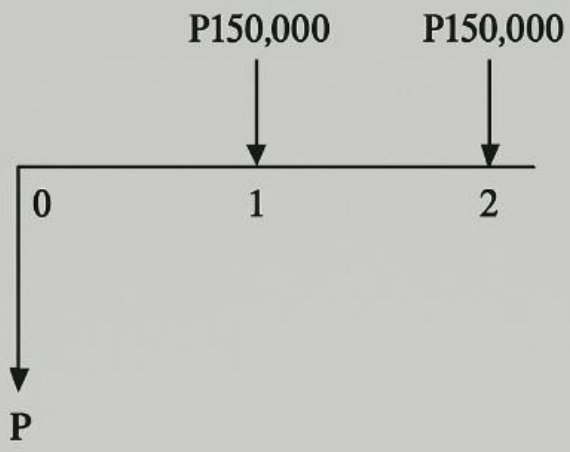
  CC = FC + PW of perpetual operation and maintenance.

Ex1: Determine the CC of an edifice that requires an initial cash outlay of ₱1.5m and an annual maintenance of ₱150k. Given interest is 15%. Ans. ₱2.5m.

CAPITALIZED COST: Case 1

Ex1: Determine the CC of an edifice that requires an initial cash outlay of ₱1.5m and an annual maintenance of ₱150k. Given interest is 15%. Ans. ₱2.5m.

Solution



$$P = \frac{A}{i} = \frac{P150{,}000}{0.15} = P1{,}000{,}000$$

Capitalized cost = First cost + P

= P1,500,000 + P1,000,000

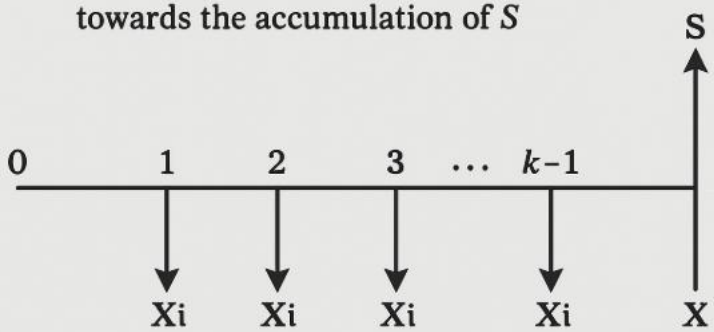
= P2,500,000

CAPITALIZED COST: Case 2

- Replacement (cost of replacement) only, no maintenance and/or operation.

  CC = FC + PW of perpetual replacement

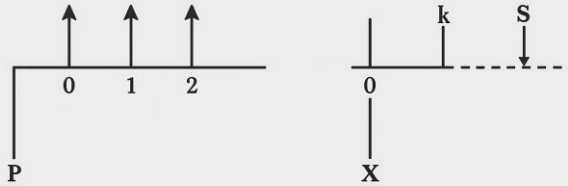
S = = amount needed to replace a property every k periods
X = amount of principal invested at rate i% the interest on which will amount to S every k periods
Xi = interest on X every period, the periodic deposit towards the accumulation of S



$$S = Xi\,(F/A, i\%, k)$$

$$X = S\left[\frac{1}{S/A, i\%\ k}\right]\frac{S}{(1-i)^k-1)}$$

Difference between P and X in a perpetuity (2–33)

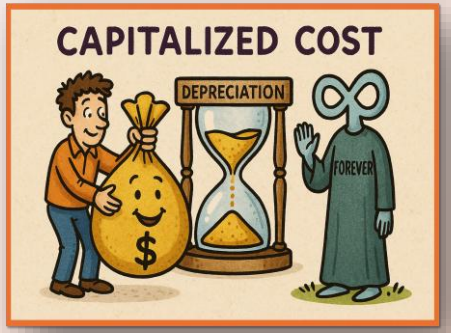


$$P = \frac{A}{i}$$

$$X = \frac{S}{(1+i)^k-1}$$

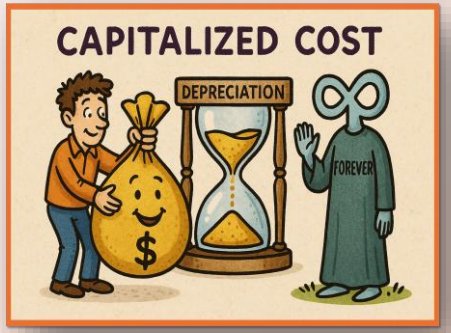
Note: if k = 1, then X = P.

CAPITALIZED COST: Case 2



Ex2: A new car engine was installed at an initial cost of ₱300k with a projected useful life of 15 years. The salvage value is ₱30k at the end of its useful life. Find the CC if interest is pegged at 18% compounded yearly. Ans. ₱324,604.

CAPITALIZED COST: Case 2



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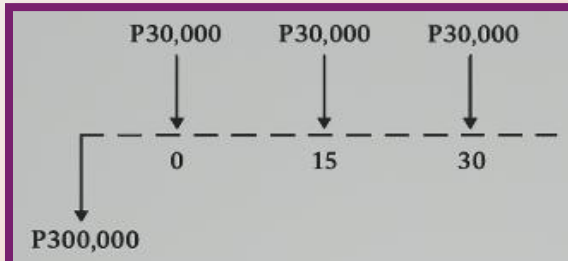
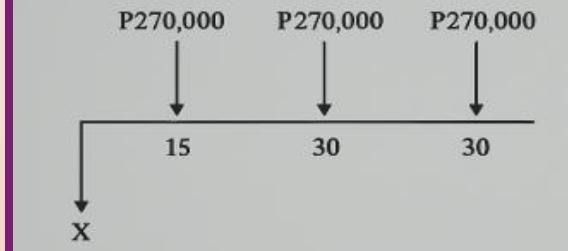


Cash flow diagram for the engine



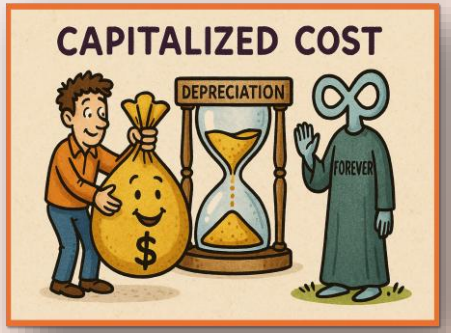
$$X = \frac{S}{(1+i)^k - 1} = \frac{P270{,}000}{(1+0{,}18)^{15} - 1} = P24{,}604$$

Capitalized cost = First cost + X = P300,000

= P300,000 + P24,604

= P324,604

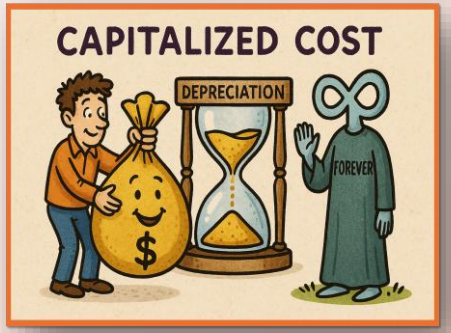
CAPITALIZED COST: Case 3



- Replacement, maintenance and/or operation every period

  CC = FC + PW of perpetual operation and/or maintenance + PW of cost of perpetual replacement

CAPITALIZED COST: Case 3

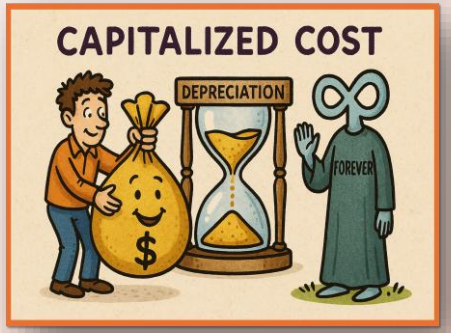


Ex3: Look for the capitalized cost of a lab site that requires:

- ₱5m for construction
- ₱0.1m at the end of every year for the 1st 6 years and ₱0.120m each year thereafter for operating expenses
- ₱0.5m every 5 years for replacement of equipment
- Interest at 12% per annum.

Ans. ₱6,753,650.

CAPITALIZED COST: Case 3



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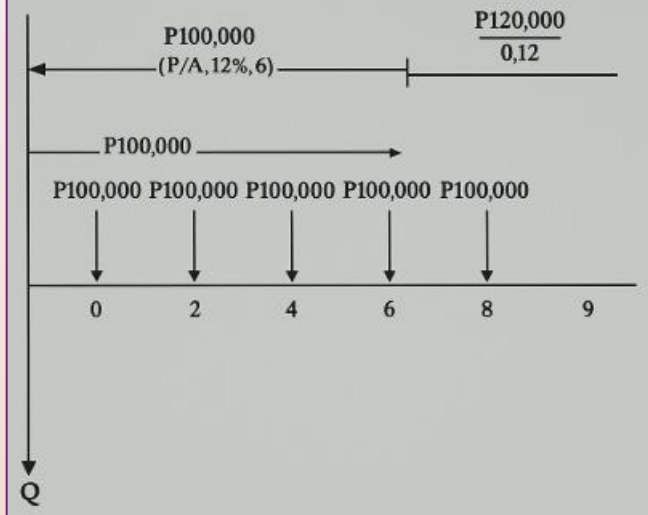
Operation:



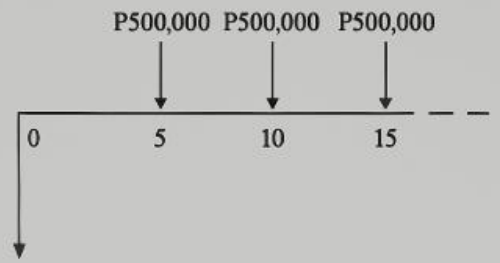
Let Q = the present worth of cost of perpetual operation

$$Q = P100{,}000\,(P/A, 12\%, 6) + \frac{P120{,}000}{0{,}12}$$

$$= P100{,}000\,(4{,}1114) + \frac{P120{,}000}{0{,}12}\,(0{,}5066)$$

$$= P917{,}740$$

Replacement:



Let X = the present worth of cost of perpetual replacement

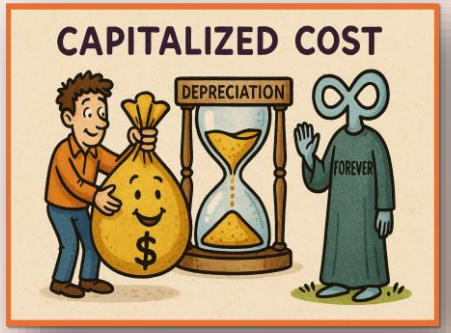
$$X = \frac{S}{(1+i)^k - 1} = \frac{P500{,}000}{(1+0.12)^5 - 1} = \mathbf{P655{,}910}$$

Capitalized cost = First cost + $Q + X$

= P5,000,000 + P917,740 + P655,910

= P6,753,650

CAPITALIZED COST: Summary

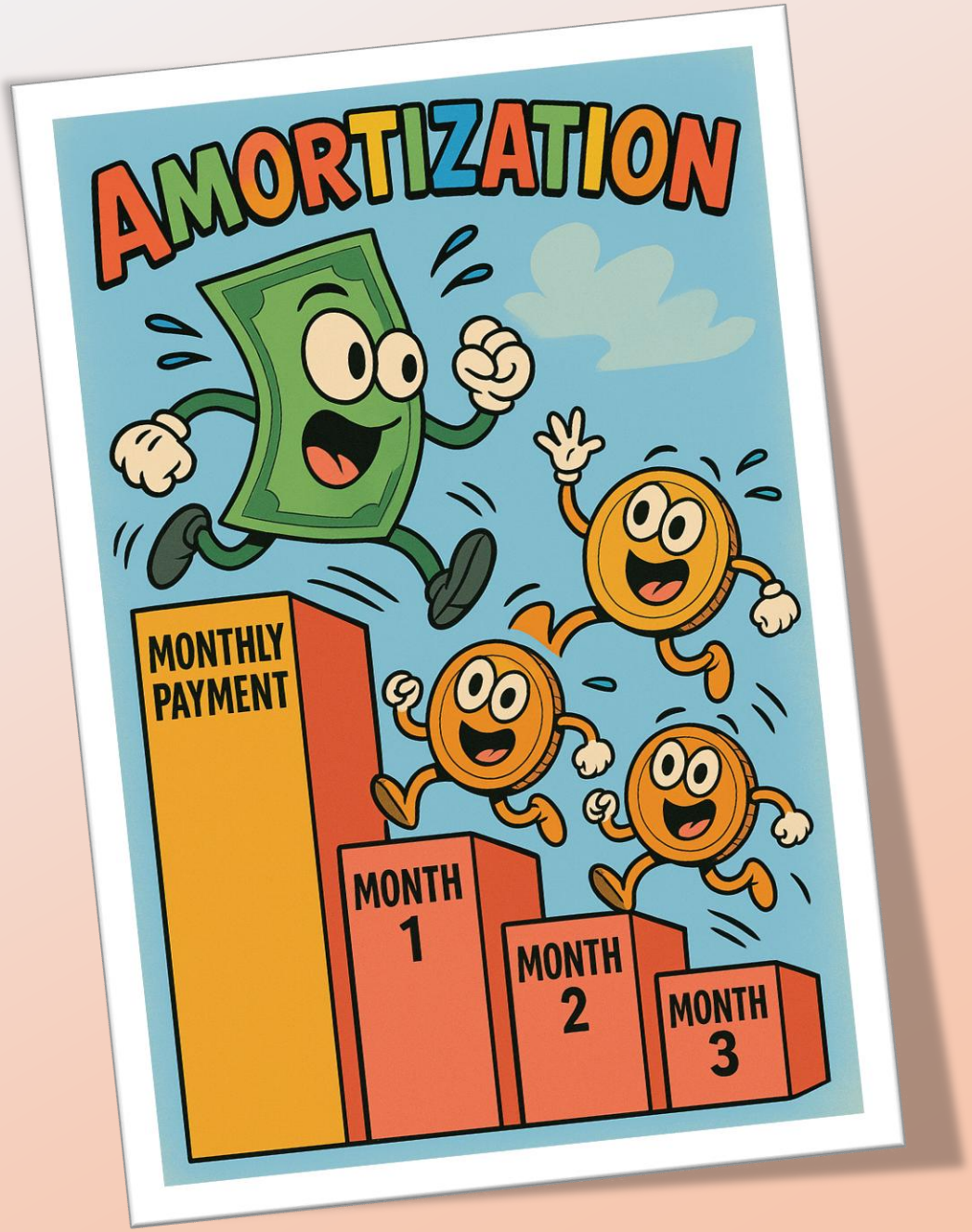


Case	Features	Formula
1	FC + PW of perpetual O&M	$\mathbf{P} = \frac{A}{i}$ (A)
2	FC + PW of perpetual replacement	$\mathbf{X} = \frac{S}{(1+i)^k - 1}$ (B)
3	FC + PW of perpetual O&M + PW of perpetual replacement	(A) + (B)

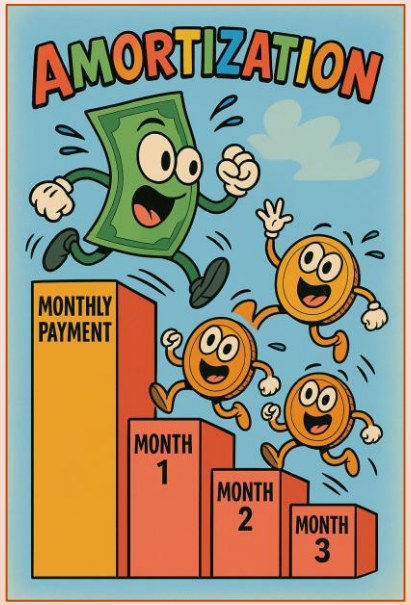
+ FC

AMORTIZATION

- Any method or repaying a debt.
- Included the principal and interest.
- Usually, a series of equal payments at equal interval of time.



AMORTIZATION



Ex1. Somebody owes ₱5k with interest at 8% compounded semi-annually. This is to be amortized by an equal semi-annual payments over the succeeding 4 years. The first due in 18th month. Find the semi-annual payment and construct an amortization schedule.

AMORTIZATION

Ex1. Somebody owes ₱5k with interest at 12% compounded semi-annually. This is to be amortized by an equal semi-annual payments over the succeeding 3 years. The first due in 6 months. Find the semi-annual payment and construct an amortization schedule. Ans. ₱1,016.82.

$$\mathbf{A} = \frac{\mathrm{P}}{\mathrm{P/A,\ 6\%,6}} = \frac{\mathrm{P5{,}000}}{4.9173} = \mathrm{P1{,}016.82}$$

Period	Outstanding principal at beginning of period	Interest due at end of period	Payment (Annuity)	Principal repaid at end of period
1	₱5,000.00	₱300.00	₱1,016.82	₱716.82
2	₱4,283.18	₱256.99	₱1,016.82	₱759.83
3	₱3,523.35	₱211.40	₱1,016.82	₱805.42
4	₱2,717.93	₱163.08	₱1,016.82	₱853.74
5	₱1,864.19	₱111.85	₱1,016.82	₱904.97
6	₱959.22	₱57.55	₱1,016.82	₱959.22

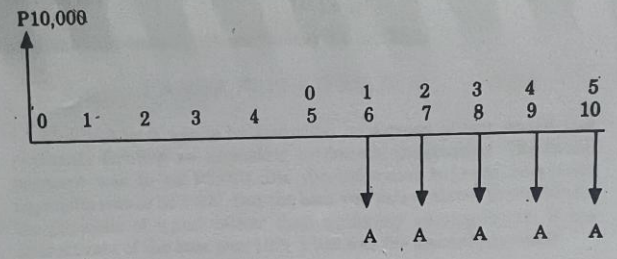
AMORTIZATION (Do!)

Ex2. A loan of ₱10,000 with interest rate of 20% SA is to amortized by 5 equal payments at the end of each of 6 months. The first payment is to be made after 3 years. Find the SA payment and construct the amortization schedule. Ans. ₱4,248.50.

AMORTIZATION (Do!)

Ex2. A loan of ₱10,000 with interest rate of 20% SA is to amortized by 5 equal payments at the end of each of 6 months. The first payment is to be made after 3 years. Find the SA payment and construct the amortization schedule. Ans. ₱4,248.50.

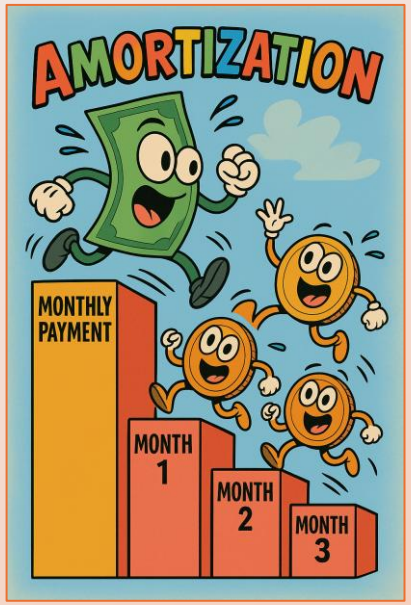
Solution



$$P = A(P/A, 10\%, 5)(P/F, 10\%, 5)$$
$$A = P(A/P, 10\%, 5)(F/P, 10\%, 5)$$
$$= P10{,}000\,(0.2638)\,(1.6105)$$
$$= P4{,}248.50$$

Amortization Schedule

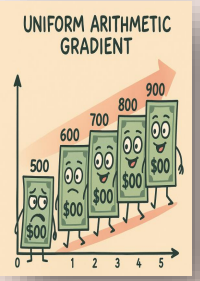
Period	Outstanding principal at beginning of period	Interest due at end of period	Payment	Principal repaid at end of period
1	P10,000.00	P 1,000.00		
2	11,000.00	1,100.00		
3	12,100.00	1,210.00		
4	13,310.00	1,331.00		
5	14,641.00	1,464.10		
6	16,105.10	1,610.51	P 4,248.50	P 2,637.99
7	13,467.11	1,346.71	4,248.50	2,901.79
8	10,565.32	1,056.53	4,248.50	3,191.97
9	7,373.35	737.34	4,248.50	3,511.16
10	3,862.19	386.22	4,248.50	3,862.28
TOTALS		P11,242.41	P21,242.50	P16,105.19



UNIFORM ARITHMETIC GRADENT

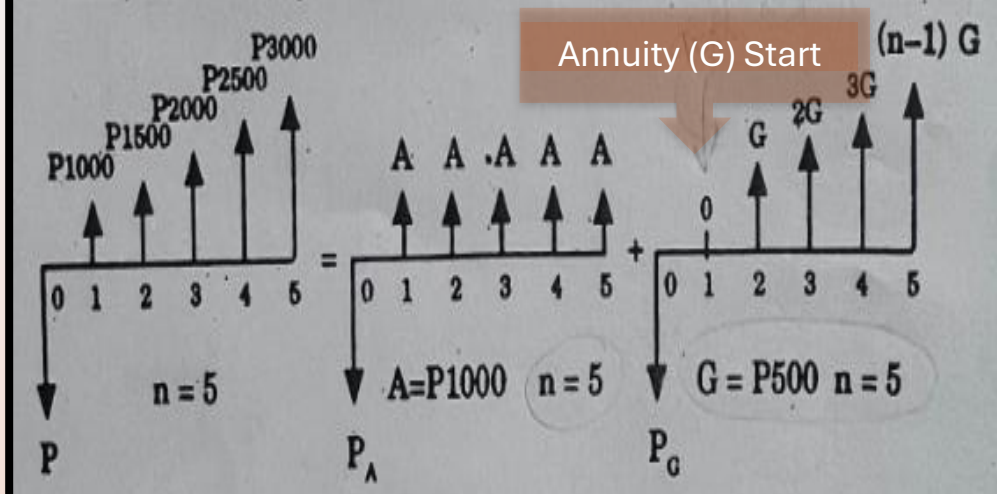
- Economic analysis problems involve receipts or dibursements that increase/decrease by a uniform amount per period.
- Examples: O&M, repair etc. may increase by a relative **constant amount** each period known as UAG.

$$\mathbf{P} = A(P/A, 1\%, n) + G\,(P/G, 1\%, n)$$



Gradient to Present Worth Conversion Factor

This cash flow may be resolved into two components



G is known as the uniform gradient amount

$$P = P_A + P_G \qquad (2\text{-}34)$$

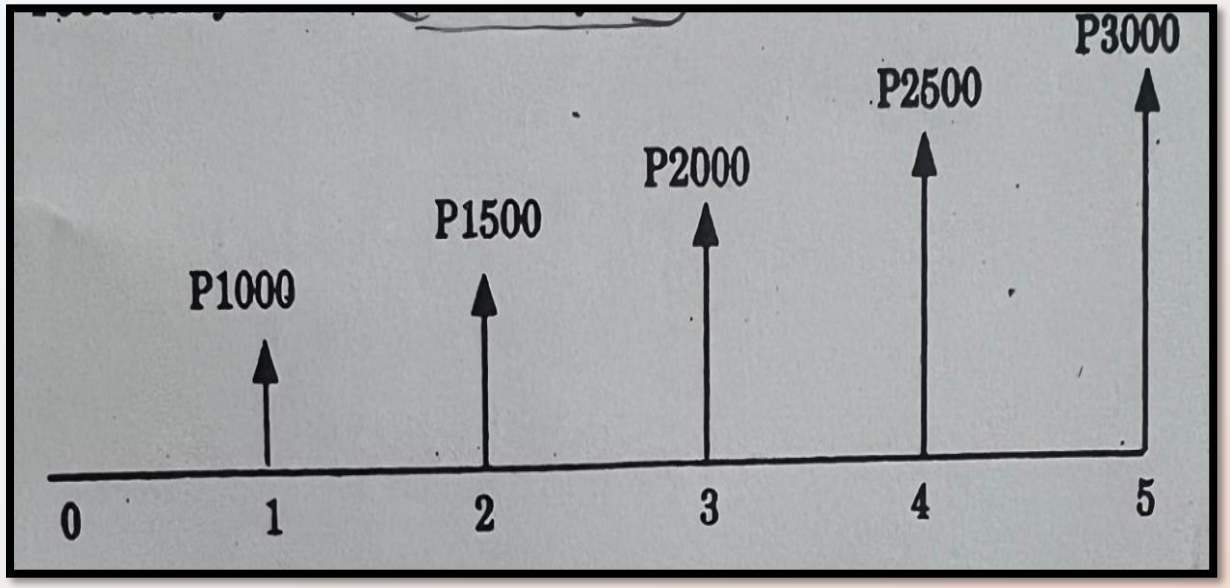
$$P_A = A(P/A, i\%, n) \qquad (2\text{-}35)$$

$$P_G = G(1+i)^{-2} + 2G(1+i)^{-3} + 3G(1+i)^{-4} + \ldots + (n-1)\,G(1+i)^{-(n-1)}$$

$$P_G = \frac{G}{i}\left[\frac{(1+i)^n - 1}{i} - n\right]\left[\frac{1}{(1+i)^n}\right] \qquad (2\text{-}36)$$

$$P_G = G(P/G, i\%, n) \qquad (2\text{-}37)$$

UNIFORM ARITHMETIC GRADENT



$$P = P_A + P_G \quad (2\text{-}34)$$

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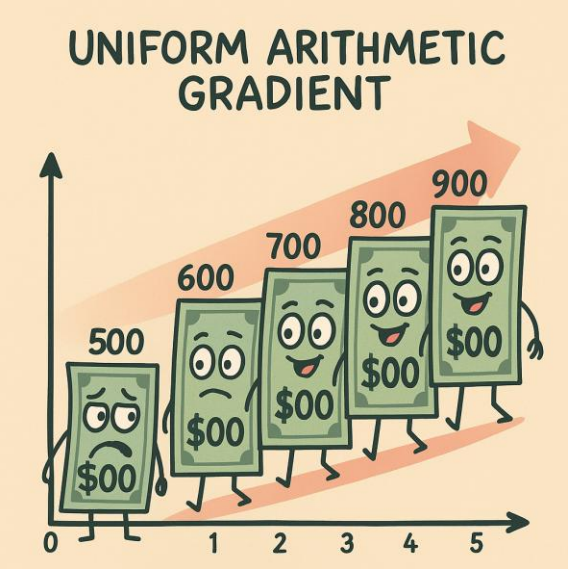
$$P_G = \frac{G}{i}\left[\frac{(1+i)^n - 1}{i} - n\right]\left[\frac{1}{(1+i)^n}\right] \quad (2\text{-}36)$$

$$P_G = G(P/G, i\%, n) \quad (2\text{-}37)$$

G = P500
n = 5
A = 1000
i = 8% SA effective

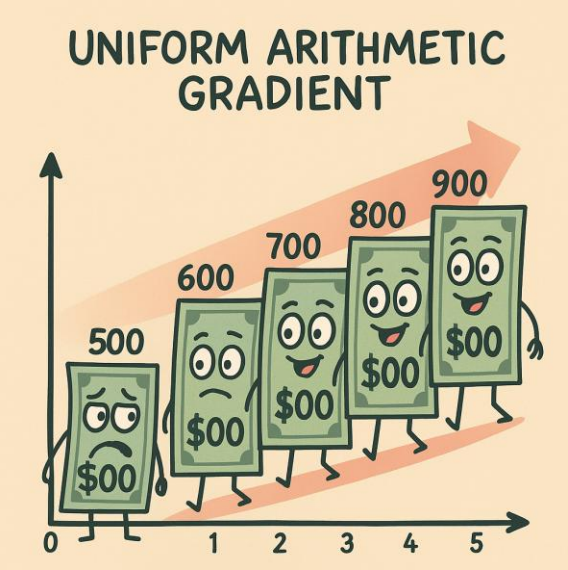
$$\mathbf{P} = A(P/A, 1\%, n) + G\,(P/G, 1\%, n)$$

UNIFORM ARITHMETIC GRADIENT



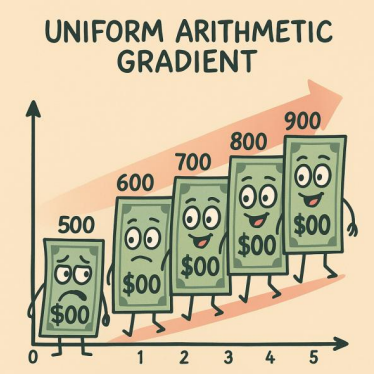
Ex1. The maintenance expense of a certain office tool is ₱1k on year 1. Afterwards, the cost increases to ₱ 500 each year for the succeeding 4 years. Find the present worth. Ans.₱

UNIFORM ARITHMETIC GRADIENT



Ex2. A loan was to be amortized by 4 end-of year ascending payments. Initial payment is ₱5k with common difference between successive payments to be ₱400. But loan was renegotiated to provide for equal payments vs. uniform varying sums. If the interest rate of loan was 15%, determine the annual payment. Ans. ₱5,530.51.

UNIFORM ARITHMETIC GRADIENT



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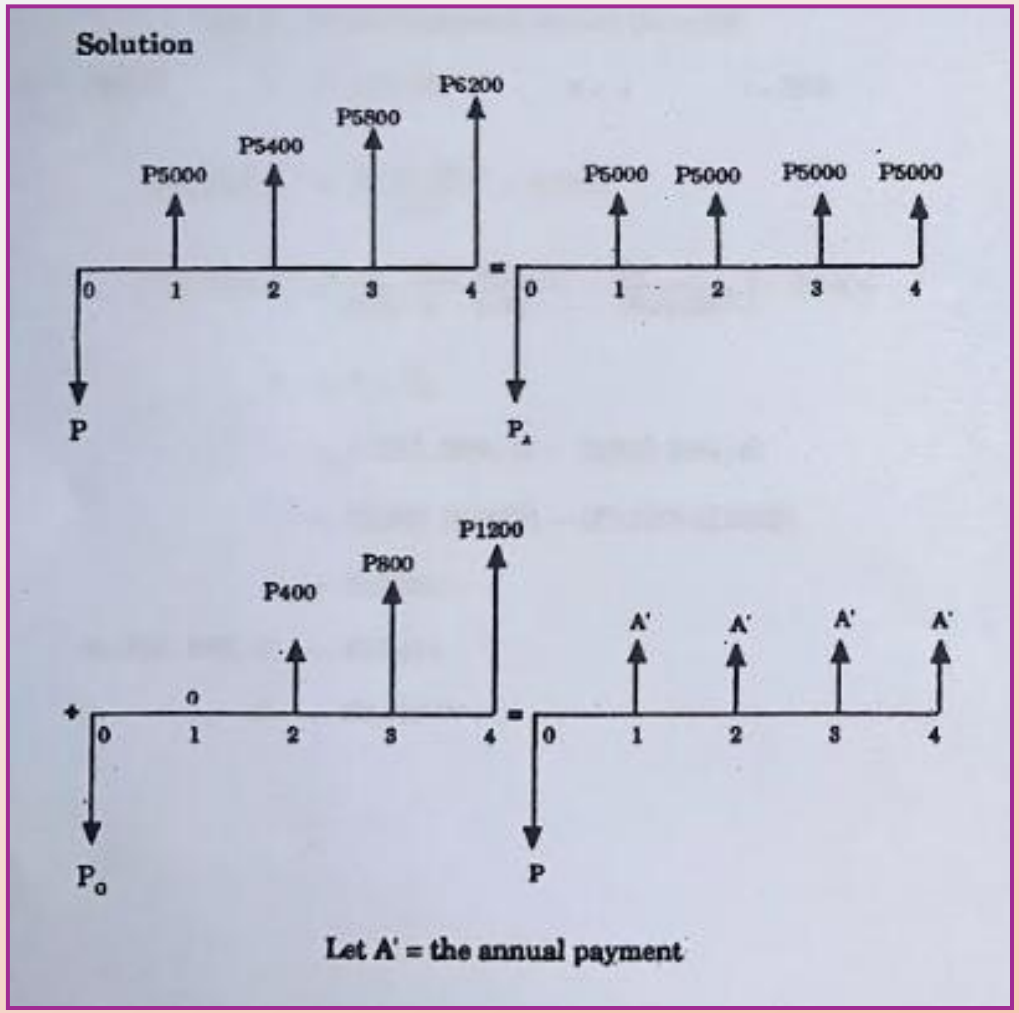


$A = P5000 \quad G = P400 \quad n = 4 \quad i = 15\%$

$$P/A, 15\%, 4 = \frac{1-(1.15)^{-4}}{0.15} = 2.8550$$

$$P/G, 15\%, 4 = \frac{1}{0.15}\left[\frac{(1.15)^4-1}{0.15} - 4\right]\left[\frac{1}{(1.15)^4}\right] = 3.7865$$

$$P = A(P/A, 15\%, 4) + G(P/G, 15\%, 4)$$

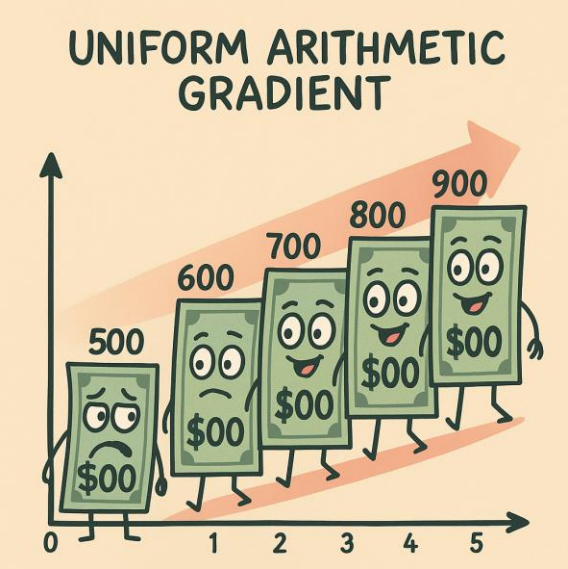
$$= (P5000)(2.8550) + P400(3.7865)$$

$$= P15{,}789.60$$

$$A'(P/A, 15\%, 4) = P15{,}789.60$$

$$A = P5530.51$$

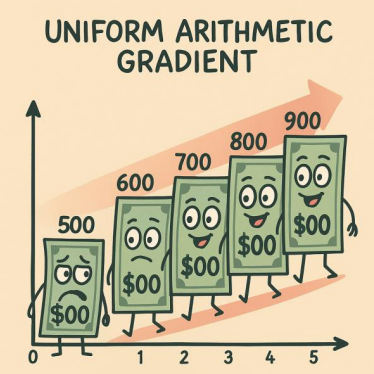
UNIFORM ARITHMETIC GRADIENT (DO!)



Ex2. Find the equivalent annual payment of the following obligations at 20% interest. Ans. ₱6,725.77.

End of Year	Payment
1	₱8,000
2	₱7,000
3	₱6,000
4	₱5,000

UNIFORM ARITHMETIC GRADIENT (Do!)



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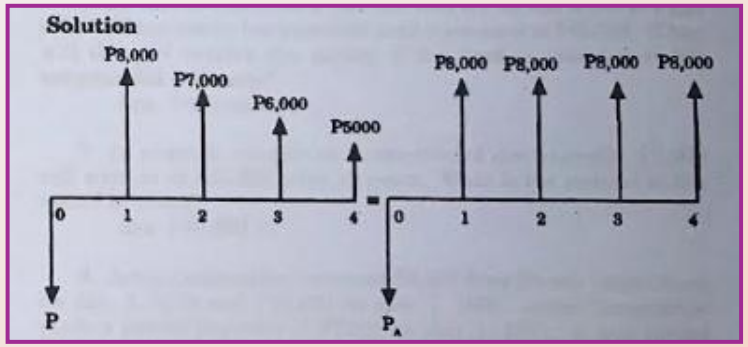
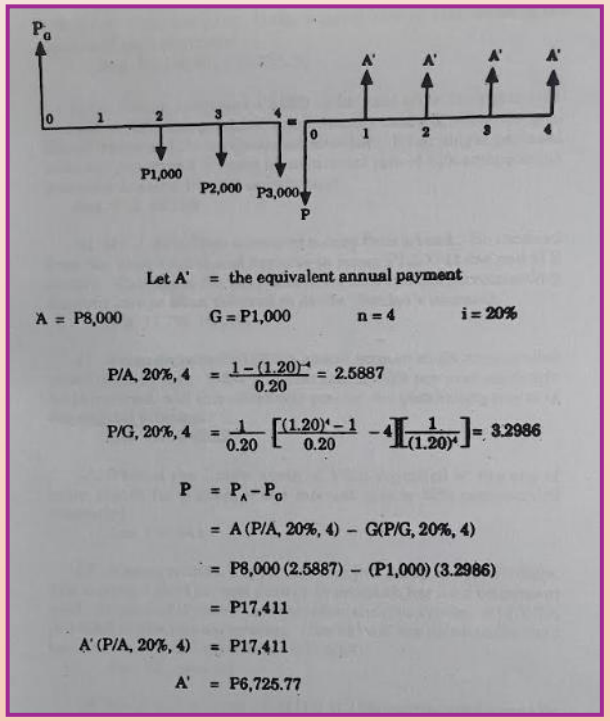
4	₱5,000





Let A' = the equivalent annual payment

A = P8,000 G = P1,000 n = 4 i = 20%

$$P/A, 20\%, 4 = \frac{1-(1.20)^{-4}}{0.20} = 2.5887$$

$$P/G, 20\%, 4 = \frac{1}{0.20}\left[\frac{(1.20)^4-1}{0.20} - 4\right]\left[\frac{1}{(1.20)^4}\right] = 3.2986$$

$$P = P_A - P_G$$

$$= A(P/A, 20\%, 4) - G(P/G, 20\%, 4)$$

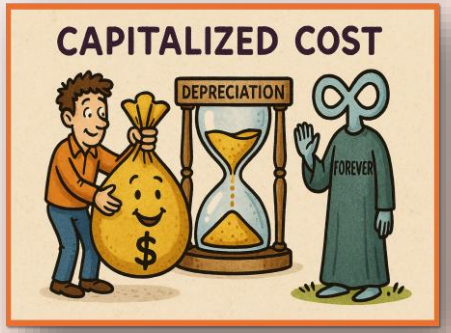
$$= P8{,}000\,(2.5887) - (P1{,}000)\,(3.2986)$$

$$= P17{,}411$$

$$A'(P/A, 20\%, 4) = P17{,}411$$

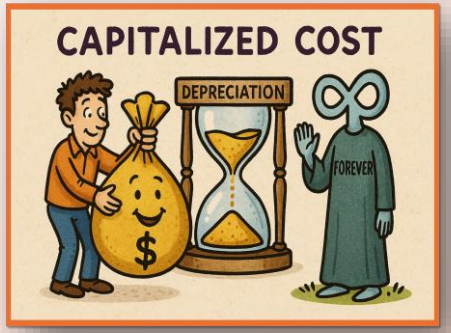
$$A' = P6{,}725.77$$

DEPRECIATION



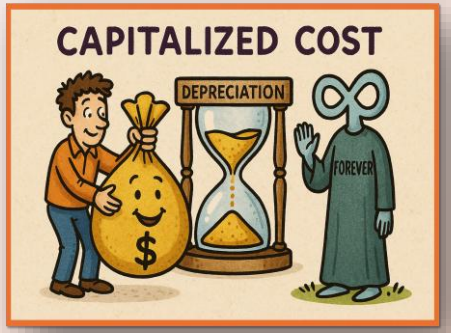
- Decrease in the value of physical property due to time.
- Concept of Value
  - **Value** = PW of all future profits to be received through ownership of a particular property.
  - **Market Value of a property** = amount which a willing buyer to pay to a willing seller for the property (but not coerced to buy the product.
  - **Utility or Use of Value of a property** = what the property is worth to the owner as an operating unit.
  - **Fair Value** = value which is usually determined by a disinterested party to establish a fair price.
  - **Book Value (Depreciated Value)** = worth of a property as shown in the books (accounting records) of an enterprise.
  - **Salvage Value (Resale Value)** = price that can be obtained from the sale of property after it has been utilized.
  - **Scrap Value** = amount of property to sell for if disposed.

DEPRECIATION



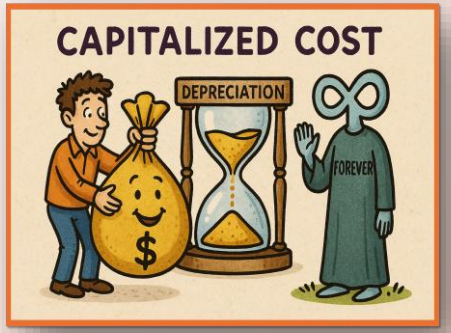
- **Purpose of Depreciation**
  - To provide for the recovery of the capital which has been invested on physical property.
  - To enable the cost of depreciation to be charged to the cost of producing products or services that results from the use of property.
- **Types of Depreciation**
  - Normal depreciation: physical or functional
  - Depreciation due to changes in price levels
  - Depletion

DEPRECIATION



- **Requirements of a Depreciation Method**
  - Simple
  - Recover Capital
  - Book Value = reasonably close to the market value at any time.
  - Accepted by BIR.

DEPRECIATION



Depreciation Methods

- The Straight Line Method
- The Sinking Fund Formula
- Declining Balance Method
- Double Declining Balance (DDB) Method
- The Sum-of-The Years’-Digits (SYD) Method

DEPRECIATION

Depreciation Method Parameters

- L= useful life of the property in years
- C_o= the original cost
- C_L= the value at the end of life - Scrap Value, included gain/loss due to removal.
- d = the annual cost of depreciation
- C_n= the book value at the end of n years
- D_n= depreciation up to age n years

THE STRAIGHT LINE METHOD

Method assume that loss in value is directly proportional to the age of property.

$$d = \frac{C_o - C_L}{L}$$

$$D_n = \frac{n(C_o - C_L)}{L}$$

$$C_n = C_o - D_n$$

An electronic equipment costs $90,000 with a salvage value of $8,000. It's end of life is 10 years. What would be the book value after 3 years, using SLM to solve for the depreciation.

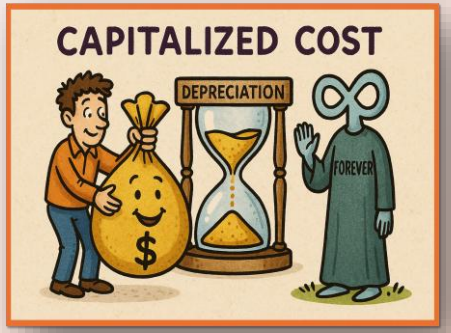
Ans.

d = $8,200

$D_3 = \$24{,}600$

$C_3 = \$65{,}400$

THE SINKING FUND FORMULA



- Sinking fund (Annuity) is established to accumulate for replacement.
- Total depreciation at any given = accumulated amount in the sinking fund at that time

$$d = \frac{C_o - C_L}{(F/A\ , i\%, L)}$$

$$C_n = C_o - D_n$$

$$D_n = d(F/A, i\%, n)$$

A telecommunications company bought an equipment for P53k and paid P1.5k plus delivery charges to the job sites. The equipment as a normal life of 10 years with a trade-in value of P5k vs the a new equipment at the end of life.

- Determine the annual depreciation cost using SLM.
- Determine the annual depreciation cost by SFM. Assume interest at 6.5% compounded annually.

Ans.

- D = P4,950
- d = P3,668

DECLINING BALANCE METHOD

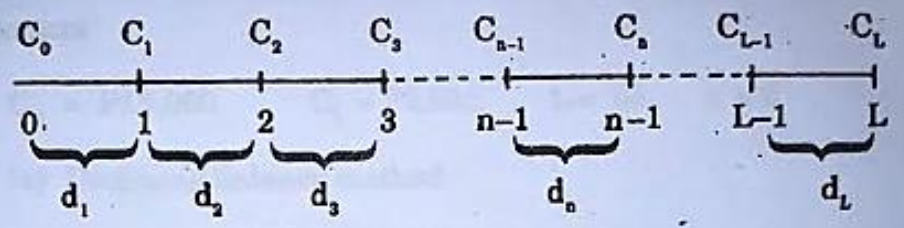
- Also called the CONSTANT PERCENTAGE METHOD or the MATHESON FORMULA.
- The Annual Cost of depreciation is a fixed percentage of the salvage value at the beginning of the year.
- The ratio of the depreciation in any year to the book value at the beginning of the year is constant throughout the life of the property, designated as *k*, the rate of depreciation

$$d_n = C_o(1-k)^{n-1}k$$

$$C_n = C_o(1-k)^n = C_o\left(\frac{C_L}{C_0}\right)^{\frac{n}{L}}$$

$$C_L = C_o(1-k)^L$$

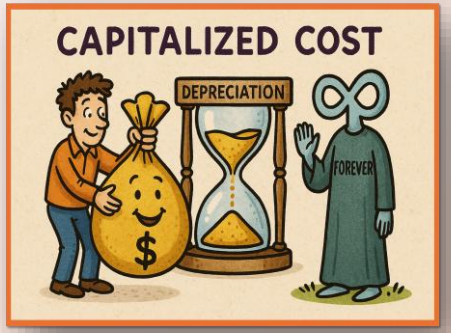
$$k = 1 - \sqrt[n]{\frac{C_n}{C_o}} = 1 - \sqrt[L]{\frac{C_L}{C_o}}$$



d_n = depreciation during the nth year

Year	Book value at beginning of year	Depreciation during the year	Book value at the end of year
1	C_0	$d_1 = kC_0$	$C_1 = C_0 - d_1 = C_0(1-k)$
2	$C_0(1-k)$	$d_2 = kC_1$	$C_2 = C_1 - d_2 = C_0(1-k)^2$
3	$C_0(1-k)^2$	$d_3 = kC_2$	$C_3 = C_2 - d_3 = C_0(1-k)^3$
--	--	--	--
n	$C_0(1-k)^{n-1}$	$d_n = kC_{n-1}$	$C_n = C_{n-1} - d_n = C_0(1-k)^n$
--	--	--	--
L	$C_0(1-k)^{L-1}$	$d_L = kC_{L-1}$	$C_L = C_{L-1} - d_L = C_0(1-k)^L$

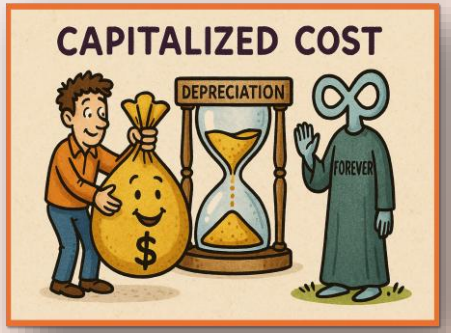
DECLINING BALANCE METHOD



Machine X loses 10% of its value yearly. A machine's original cost is P2,000. Prepare a table showing yearly depreciation, the total depreciation and the book value at the end of each year for 5 years.

Year	Book Value at the beginning of year	Depreciation during the year 10%	Total depreciation at the end of the year	Book value at the end of year
1	₱2,000.00	₱200.00	₱200.00	₱1,800.00
2	₱1,800.00	₱180.00	₱380.00	₱1,620.00
3	₱1,620.00	₱162.00	₱542.00	₱1,458.00
4	₱1,458.00	₱145.80	₱687.80	₱1,312.20
5	₱1,321.20	₱132.22	₱819.12	₱1,180.98

DOUBLE DECLINING BALANCE METHOD



Similar to declining balance method. But replace k by $\frac{2}{L}$.

$$d_n = C_o(1 - \frac{2}{L})^{n-1}\frac{2}{L}$$

$$C_n = C_o(1 - \frac{2}{L})^{n}$$

$$C_L = C_o(1 - \frac{2}{L})^{L}$$

When DDB Method is used, the salvage value should not be subtracted from the first cost when calculating the depreciate charge.

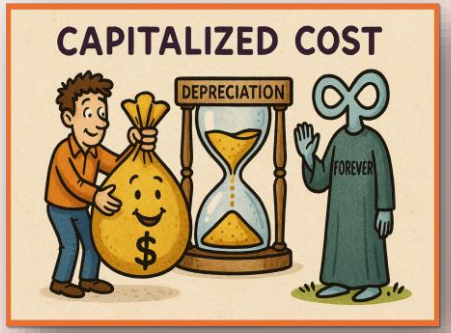
Determine the rate of depreciation, the total depreciation at the end of the 8th year, the book value at the end of the 8th years for an asset that costs P15,000 for a new item and has an estimated scrap value of P2,000 at the end of 10 years using:

(a) DBM
(b) DDBM

a) DBM
- $k = 0.1825 = 18.25\%$
- C_8= P2,992
- D_8= P12,008

b) DDBM
- $\frac{2}{L} = 0.2 = 20\%$
- C_8= P2,517
- D_8= P12,483

THE SUM-OF-THE-YEAR'S DIGITS (SYD) METHOD



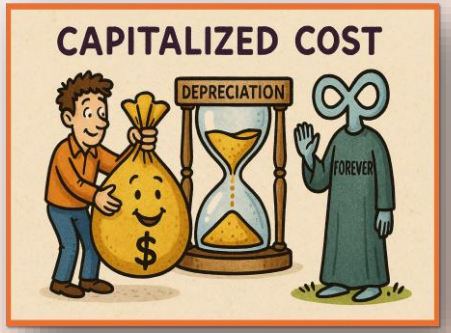
$$d_n = \frac{reverse\ digit}{sum\ of\ digits}(C_o - C_L)$$

A certain new structure costs P12,000 with a useful life of 5 years & a salvage value of P1,000 at the end of life. Prepare a table to show the book value at the end of each year life.

Year	Year in Reverse Order	Depreciation during the year	Book Value at end of year
1	5	₱3,666.67	₱8,333.33
2	4	₱2,933.33	₱5,400.00
3	3	₱2,200.00	₱3,200.00
4	2	₱1,466.67	₱1,733.33
5	1	₱733.33	₱1,000.00

Formula for sum of consecutive digits = $\frac{n}{2}(\mathrm{n}+1)$

PUTTING IT ALL TOGETHER!



- Cost of machine: P7,000.
- Useful life = 8 years
- Salvage value at end of life = P350

Compute for the depreciate charge and book value at 4th & 7th year using:

1. SLM
2. SFM at 12% interest
3. DBM
4. DDBM

PUTTING IT ALL TOGETHER!

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1. SLM
2. SFM at 12% interest
3. DBM
4. DDBM

(a) Straight line method

$$d = \frac{C_o - C_L}{L} = \frac{P7{,}000 - P350}{8} = P831$$

$$C_4 = C_o - D_4 = P7{,}000 - P831(4) = P3{,}676$$

(b) Declining balance method

$$k = 1 - \sqrt[L]{\frac{C_L}{C_o}} = 1 - \sqrt[8]{\frac{P350}{P7{,}000}} = 0.3123$$

$$d_4 = C_o(1-k)^{4-1}\,k = P7{,}000\,(1-0.3123)^3\,(0.3123) = P711$$

$$C_4 = C_o(1-k)^4 = P7{,}000\,(1-0.3123)^4 = P1{,}566$$

(c) SYD method

1	8
2	7
3	6
4	5
5	26
6	
7	
8	
36	

$$\text{Sum of digits} = \frac{8(8+1)}{2} = 36$$

$$d_4 = (5/36)\,(P7{,}000 - P350) = P924$$

$$D_4 = (26/36)\,(P7{,}000 - P350) = P4{,}803$$

$$C_4 = C_o - D_4 = P7{,}000 - P4{,}803 = P2{,}197$$

(d) Sinking fund method

$$d = \frac{C_o - C_L}{F/A,\ 12\%,\ 8} = \frac{P7{,}000 - P350}{12.2997} = P541$$

$$D_4 = d(F/A,\ 12\%,\ 4) = P541\,(4.7793) = P2{,}586$$

$$C_4 = C_o - D_4 = P7{,}000 - P2{,}586 = P4{,}414$$

PROBLEM SETS.

PROBLEMS

1. A machine shop purchased 10 years ago a milling machine for P60,000. A straight-line depreciation reserve had been provided on a 20 year life of the machine. The owner of the machine shop desires to replace the old milling machine with a modern unit of many advantages costing P100,000. It can sell the old unit for P20,000. How much new capital will be required for the purchase?

Ans. P50,000

2. A tax and duty free importation of a 30 HP sandmill (for paint manufacturing) cost P360,000, CIF Manila. Bank charges, arrastre and brokerage cost P5,000. Foundation and installation costs were P25,000. Other incidental expenses amounted to P20,000. Salvage value of the mill is estimated to be P60,000 after 20 years. Find the appraisal value of the mill, using straight-line depreciation, at the end of (a) 10 years and (b) 15 years.

Ans. (a) P235,000 (b) P147,500

3. On January 1, 1978, the purchasing engineer of a Cement Co. purchased a new machine at a cost of P140,000. Depreciation has been computed by the straight-line method based on an estimated useful life of five years and residual scrap value of P12,800. On January 2, 1981, extraordinary repairs (which were almost equivalent to a rebuilding of the machinery) were performed at a cost of P30,400. Because of the thorough going nature of these repairs, the normal life of the machinery was extended materially. The revised estimate of useful life was four years from January 1, 1981.

Determine the annual provisions for depreciation for the years 1978 to 1980 and the adjusted provision for depreciation on December 31, 1981. Assume payment in cash for the machine and extraordinary repairs.

Ans. P25,440; P20,320

4. Power to a remote transmitting station is provided by a Diesel-electric generator unit. The original cost of the unit is P65,000. It costs P2,000 to ship the unit to the job site. An additional cost of P3,000 was incurred for installation.

(a) Determine, the annual depreciation cost by the straight line method, if the unit has an expected life of 10 years. The salvage value of the unit at the end of its life was estimated at P5,000.

(b) Determine the annual depreciation cost by the sinking fund method. Assume that the annual charge for depreciation was deposited in a fund drawing compound interest at the rate of 5%

Ans. (a) P6,500 (b) P5,168

5. An industrial plant bought a generator set for P90,000. Other expenses including installation amounted to P10,000. The generator set is to have a life of 17 years with a salvage value at the end of life of P5,000. Determine the depreciation charge during the 13th year and the book value at the end of 13 years by the (a) declining balance method, (b) double declining balance method, (c) sinking fund method at 12% and (d) SYD method.

Ans. (a) P1,949; P10,118 (b) P2,620; P19,649 (c) P1,943; P45,539 (d) P3,105; P11,209

6. A telephone company purchased a microwave radio equipment for P6,000,000. Freight and installation charges amounted to 3% of the purchase price. If the equipment shall be depreciated over a period of 8 years with a salvage value of 5%, determine the following:

(a) Annual depreciation charge using the straight line method.

(b) Depreciation charge during the fifth year using the sum-of-the-years-digits method.

Ans. (a) P733,875 (b) P652,333